The Case for Voluntary Disclosure

Strategic Feature Revelation in Regression & Classification

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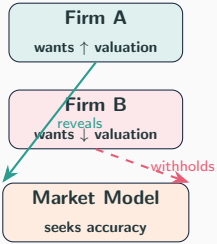
Strategic Learning with Heterogeneous Preferences

Motivation & Core Question

Setting: A predictive model relies on features from strategic participants.

- **Mandatory:** All features disclosed $\Rightarrow$ baseline accuracy
- **Voluntary:** Participants *choose* what to reveal based on private **preferred outcomes**

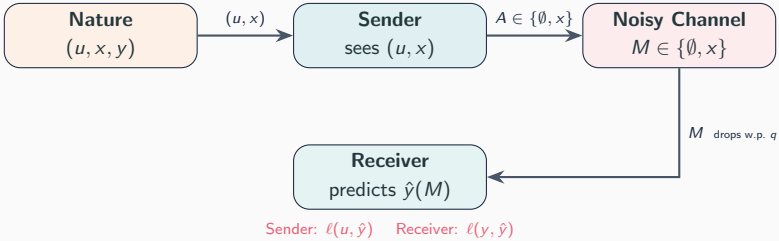
Example: Firms in financial markets selectively disclose financials to influence their valuation—some want $\uparrow$, others want $\downarrow$.



Central Question

When does voluntary disclosure actually **improve** *the model's performance?*

Game-Theoretic Model



Stackelberg game: Receiver commits $\hat{y}$ $\rightarrow$ Sender best-responds $\rightarrow$ outcome realized.

Strategic Risk (Voluntary)

$$R_S := \min_{\hat{y}} \mathbb{E}[\ell(Y, \hat{y}(M))]$$

Vanilla Risk (Mandatory)

$$R_V := \mathbb{E}[\ell(Y, \hat{y}_V(M))] \quad (A \equiv x)$$

Strategic Advantage: $R_S < R_V$ — voluntary *outperforms* mandatory

Regression: Sufficiency & Necessity

Theorem (Sufficiency)

Strategic advantage exists if $\exists\, b$ such that:

$$\|U - Y\|_2 \;<\; \frac{\mathbb{E}[|G|] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}}$$

Theorem (Necessity)

$$\max_b \left(\mathbb{E}[|G|] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X) \right) > 0$$

Interpretation:

- **LHS** = preference misalignment $\|U - Y\|_2$
- **RHS** = feature informativeness vs. sub-optimality

Both conditions are **independent of** q!

q scales magnitude, not existence.

Gaussian Specialization

When $(U, X, Y) \sim \mathcal{N}(0, \Sigma)$, the sufficiency condition has a **closed-form**:

Theorem (Gaussian Sufficiency)

With $b = 2\Sigma_{XX}^{-1}\Sigma_{XU}$, strategic advantage exists if:

$$2\left(b_Y^\top \Sigma_{XU} - \sigma_Z^2\right) + \frac{8\kappa\,\sigma_Z}{s\sqrt{2\pi}}\, e^{-\alpha^2/2} \;>\; \Sigma_{XY}^\top \Sigma_{XX}^{-1} \Sigma_{XY}$$

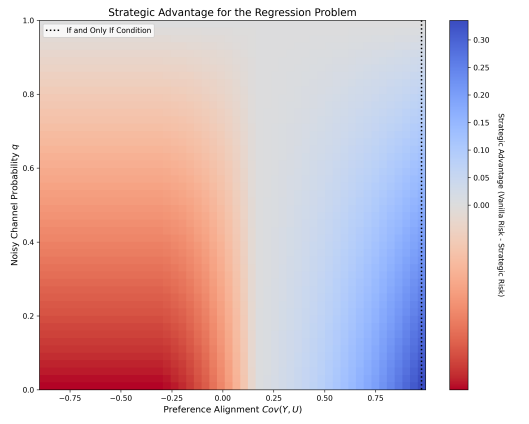
Parameters

- $b_Y = \Sigma_{XX}^{-1}\Sigma_{XY}, \;\; b_U = \Sigma_{XX}^{-1}\Sigma_{XU}$
- $\sigma_Z^2 = \Sigma_{XU}^\top \Sigma_{XX}^{-1}\Sigma_{XU}$
- $\kappa = \mathrm{Cov}(U, Y \mid X), \;\; s^2 = \mathrm{Var}(U \mid X)$

Why It Matters

- Everything is a function of Σ
- Directly **verifiable** from data
- Provides **intuitive** parameter interpretations

Regression: Numerical Validation



Setup: $(X, Y, U) \sim \mathcal{N}(0, \Sigma)$

- Unit marginal variances
- $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, U) = 0.2$
- Sweep $q \in [0, 1]$ and $\mathrm{Cov}(Y, U) \in [-1, 1]$

Result

The analytical sufficiency bound (**dotted contour**) reliably delineates the true advantage region.

Advantage region is **vertical** $\Rightarrow$ confirms independence from q.

Classification: Stackelberg Equilibrium

Case 1: $\mathcal{S}_1$ dominates $\forall q$

When $R^\star_{\mathcal{S}_1}(0) \leq R^\star_{\mathcal{S}_{-1}}(0)$:

$$(1-q)\left[\mathbb{E}\left[\frac{1-|\eta|}{2}\right] - \pi_{-1}\mathbb{E}\left[\frac{1-|\eta_{-1}|}{2}\right] - \pi_1\mathbb{P}(Y=-1 \mid U=1)\right] > 0$$

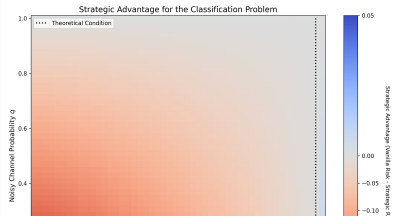
Independent of q (like regression)

Case 2: Switch at $q^\star$

$\exists\, q^\star$: use $\mathcal{S}_1$ for $q \geq q^\star$, $\mathcal{S}_{-1}$ for $q < q^\star$:

$$(1-q)\left[\mathbb{E}\left[\frac{1-|\eta|}{2}\right] - \pi_1\mathbb{E}\left[\frac{1-|\eta_1|}{2}\right]\right] + q\left[\mathbb{P}(Y=-1) - \pi_1\mathbb{P}(Y=1 \mid U=1)\right] > \pi_{-1}\mathbb{P}(Y=1 \mid U=-1)$$

Depends on q (unlike regression)



Online Learning: Unknown Environment

Challenge

Previous results assume known distributions. Now the receiver must **learn** through repeated interaction.

Bayesian Adaptive Protocol:

1. Initialize prior $p_1(\theta)$ over Θ
2. Each round t: select regime D_t, commit $\hat{y}_t$, observe y_t
3. Update: $p_{t+1}(\theta) \propto p_t(\theta) \cdot \mathbb{P}(M_t, y_t \mid \theta, \hat{y}_t)$

Regret:

$\mathcal{R}_T = \sum_t \ell(y_t, \hat{y}_t(M_t)) - T \cdot \min\{R_V^\star, R_S^\star\}$

Regret Guarantee

$$\boxed{\mathbb{E}[\mathcal{R}_T] \leq C\sqrt{T}}$$

Decomposition:

- **Estimation:** $O(\sqrt{T})$
- **Regime ID:** $O(\log T)$

$\Rightarrow$ Converges to optimal regime!

Summary & Future Directions

Regression
- Sufficiency & necessity
- Independent of q
- Gaussian closed-form

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Classification
- Full Stackelberg eq.
- Two-case structure
- q-dependent threshold

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Online Learning
- Bayesian adaptive protocol
- $O(\sqrt{T})$ regret
- Converges to optimal

Limitations

- Conservative sufficiency bounds
- Linear strategies only (regression)
- Parametric environment (online)

Future Work

- Non-linear predictors
- Multi-sender competition
- Partial feature disclosure
- Tighter bounds

Thank You!

Questions & Discussion

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Key Message

Voluntary disclosure can outperform mandatory disclosure when preferences are sufficiently aligned with ground truth.